

House Price Prediction Using Linear Regression

1. Introduction

The objective of this project is to build a machine learning model that predicts house prices using the California Housing dataset. The project demonstrates the complete workflow including data loading, exploratory data analysis (EDA), model training, evaluation, and prediction.

2. Dataset Description

The dataset contains 20,640 records with 8 input features such as Median Income, House Age, Average Rooms, Population, Latitude, and Longitude. The target variable is House Price.

3. Exploratory Data Analysis (EDA)

EDA includes checking missing values, visualizing distributions using histograms, and analyzing relationships between variables using correlation heatmaps.

4. Model Development

The dataset was split into training and testing sets (80/20). A Linear Regression model was trained using the training data.

5. Model Evaluation

Metric	Value
MAE	0.5332001304956555
MSE	0.5558915986952442
R2 Score	0.575787706032451

6. Conclusion

The Linear Regression model achieved moderate performance with an R2 score of 0.57. It provides a good baseline but can be improved using advanced models.

7. Future Improvements

Future improvements include using Random Forest, Gradient Boosting, feature engineering, and hyperparameter tuning to improve prediction accuracy.